AP CALCULUS AB

Unit 1 • Worksheet 1-5A • Answer key

Worksheet 1-5A • IVT and EVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

f is continuous on $[1,5]$ with $f(1) = -2$ and $f(5) = 6$. Which must be true?

(A) $f(c) = 8$ for some c

(B) $f(3) = 2$

(C) f is increasing

(D) $f(c) = 4$ for some c in $(1,5)$

Solution

Answer: (D). IVT.

Question 2

NO CALCULATOR

Which function is guaranteed to have an absolute maximum on the given interval?

(A) $\frac{1}{x}$ on $(0,1]$

(B) x^2 on $(-1,2)$

(C) $\sqrt{x}$ on $[0,4]$

(D) $\tan x$ on $\left[0, \frac{\pi}{2}\right)$

Solution

Answer: (C). Continuous on a closed interval (EVT).

Question 3

NO CALCULATOR

g is continuous with $g(0) = 2$, $g(1) = -1$, $g(2) = 1$, $g(3) = 4$. What is the least number of zeros of g on $[0,3]$?

(A) 0

(B) 1

(C) 3

(D) 2

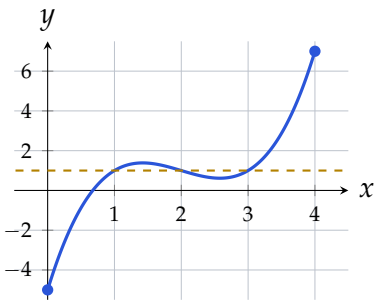
Solution

Answer: (D).

Question 4

NO CALCULATOR

The graph of $f(x) = (x-1)(x-2)(x-3)+1$ on $[0,4]$ is shown with the line $y = 1$. How many values of c in $[0,4]$ satisfy $f(c) = 1$?



(A) exactly one
(B) exactly two
(C) three
(D) none

Solution
Answer: (C). $c = 1, 2, 3$; the IVT only guarantees one.

Question 5 (Free response)

CALCULATOR

Let $f(x) = x^3 - 2x - 2$.

(a) Use the IVT to show that f has a zero in $[1,2]$.
(b) Explain why f has an absolute maximum and minimum on $[0,2]$, and use a calculator to find them.
(c) Approximate the zero in $[1,2]$ to three decimal places.

Solution
(a) Continuous (polynomial); $f(1) = -3 < 0 < 2 = f(2)$.
(b) Continuous on the closed interval $[0,2]$ (EVT). The graph's lowest point is at $x = \sqrt{2/3} \approx 0.816$: minimum ≈ -3.089; maximum $f(2) = 2$.
(c) $x \approx 1.769$.

Question 6 (Free response)

NO CALCULATOR

The temperature W (°C) in a greenhouse is continuous; $W(0) = 20$, $W(3) = 24$, $W(5) = 31$, $W(8) = 27$, $W(10) = 22$.

(a) Must there be a time when $W(t) = 30$? Justify.
(b) Explain why there are at least two times when $W(t) = 25$.
(c) Must W have an absolute maximum on $[0,10]$? Is it at least 31?

Solution
(a) Yes: continuous on $[3,5]$ and $24 < 30 < 31$ (IVT).
(b) $[3,5]$: $24 < 25 < 31$; $[8,10]$: $22 < 25 < 27$; two separate intervals.
(c) Yes (EVT: continuous on a closed interval). Yes, since $W(5) = 31$.